



New rules limit free expression

New social media rules threaten the freedom of free expression in India. <http://digit.in/mar21-49>



Parental controls for YouTube

A feature called "supervised experiences" will allow parents to restrict age-appropriate content for kids. <http://digit.in/mar21-50>

IoT security like no other

Micron's Authentica is all about tackling IoT security down to the deepest hardware level

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Micron recently unveiled their partnership with TATA Communications in the country to accelerate IoT deployment leveraging Authentica, an IoT security solution that provides protection for the lowest layers of Internet of Things (IoT) device software. Authentica allows developers to strengthen security using standard flash memory sockets without any additional hardware, thus, making their solution more affordable without compromising on the level of security. Couple that with their cloud security solution and you have zero-touch onboarding with the ability to modify the configuration in the future via the cloud. We spoke with Luis Ancajas, Director IoT Security Software Solutions at Micron Technology to know more.

▶ A lot of the IoT clients generally prefer some sort of customisation with the solution that they procure. When you bake it onto the silicon, what is the flexibility of customising that solution? How much can you offer your clients versus, let's say, a company that has an authentication system that requires the swappable security module instead of a cloud-based solution like yours.

Luis: Wild things can be baked in the Silicon, they can also be programmable with some flexibility to customize it. And the important customization that is needed in



Luis Ancajas, Director IoT Security Software Solutions, Micron Technology

security is the provisioning of a key. So it's a two-edged sword. If you provision a key on this at a software level, it's easy and you can do it after. However, you're not doing it safely because somebody could take the key and change it. Somebody could steal it, you know, so it's important to put keys in very important key storage areas usually buried underneath the Silicon. So, you put it inside technologies that obscure or obfuscate it. The key is the first point of trust and you give that to the customer. But our technology is programmable so you can also then administer a change of key and allow the customer to replace it. So there's that flexibility present. However, you always need that – they usually call it – birth Root of Trust (RoT). You know you have to have trust to be born somewhere so that you can then move it along and continue to trust it as it moves forward.

▶ Countries have been increasingly pushing for localisation of data. With a solution like yours which aims to remove borders and would require a centralised system. How do you ensure that your solution plays well with a country's localisation policy?

Luis: So I think the very foundation that you need when you localise it to a state or a country is that you need identifiers so that you can track and administer the machine that's talking to you, and strongly authenticate that. But there is another localisation that's occurring not just with the machines that are in datacenters but with the machines that are in the field. And you even have a much more fine-grained localisation there. And you actually then start to put some localisation storage and data there. Now these are no longer protected in the data center within a locked system where you know that you have to have privilege access but it's out in the field deployed and so security is also very important there.

▶ With IoT, there's this requirement for intercommunication between different standards. One client might be using a particular standard and you might have to ensure that their devices can talk to yours which could be using a different standard. Are you sort of using any universal communication protocol across the devices?

Luis: There are definitely different standards to communicate from machine to machine. It would be with HTTP or other other Internet standards. Then there are different standards for cryptography. So you have to follow those standards and that's really the goal. You start off with some basic hardware foundation and you have to put the appropriate layers to work with the public key infrastructure. Those are all standard cryptography and you don't want to change that. What you do want to have is the ability to do that in the devices. And that's also the harder thing. Following the standards is something that is done all the time, but being able to do that you have to have the right capability in your systems. ▶